

OpenCTI

Cyber Threat Intelligence Report



Cyber Threat Intelligence (CTI)

Cyber Threat Intelligence (CTI) is the process of collecting, analyzing, and using information about cyber threats. It helps organizations understand:

- Who is attacking (Threat Actors)
- How they attack (Tactics, Techniques, Procedures – TTPs)
- What tools they use (Malware, Infrastructure)

It turns raw technical data (IPs, domains, malware hashes) into actionable knowledge. And improves detection and prevention of cyberattacks. Also helps incident response teams quickly identify and contain threats.

OpenCTI

OpenCTI takes this a step further by providing a powerful platform to organize, analyze, and visualize CTI. It helps organizations not only to see the data, but also understand the story behind an attack.

- Provides a graph-based view of relationships:
Threat Actor → Campaign → Malware → Infrastructure
- Centralizes threat intelligence in one place.
- Integrates with MISP, SIEMs (Splunk, Wazuh, ELK), and case management platforms (TheHive).

Benefits:

- Visualize the full attack scenario instead of isolated IoCs.
- Link technical data (IoCs) with higher-level intelligence (actors, campaigns).
- Enhance SOC, Threat Hunting, and Incident Response operations.
- Enable collaboration between teams and organizations.

Environment Setup

Use Ubuntu Vm's terminal:

1. System update:

```
sudo apt update && sudo apt upgrade -y
```

```
mehrab@mehrab-virtual-machine:/$ sudo apt update && sudo apt upgrade -y
Hit:1 http://pk.archive.ubuntu.com/ubuntu jammy InRelease
```

2. Docker aur docker-compose install:

```
sudo apt install -y docker.io docker-compose
```

```
mehrab@mehrab-virtual-machine:/$ sudo apt install -y docker.io docker-compose
Reading package lists... Done
Building dependency tree... Done
```

3. OpenCTI repo clone:

```
git clone https://github.com/OpenCTI-Platform/docker.git opencti-docker
```

```
cd docker
```

4. Generate token:

```
uuidgen
```

```
mehrab@mehrab-virtual-machine:~/docker$ uuidgen
20d84f63-b2ab-45ee-a227-47c110b0875f
```

5. Env setup:

```
mehrab@mehrab-virtual-machine:~/docker$ ls -a
.          docker-compose.yml  .git          README.md
..         .env                 .gitignore   renovate.json
docker-compose.dev.yml  .env.sample     rabbitmq.conf
```

```
cp .env.sample .env
```

Paste generated Key in .env file.

```
GNU nano 6.2 .env
OPENCTI_ADMIN_EMAIL=admin@opencti.io
OPENCTI_ADMIN_PASSWORD=changeme
OPENCTI_ADMIN_TOKEN=20d84f63-b2ab-45ee-a227-47c110b0875f
OPENCTI_BASE_URL=http://localhost:8080
OPENCTI_HEALTHCHECK_ACCESS_KEY=changeme
```

Save Chanings.

6. Start container:

Use: `docker compose up -d`

Use url in browser to access OpenCTI.

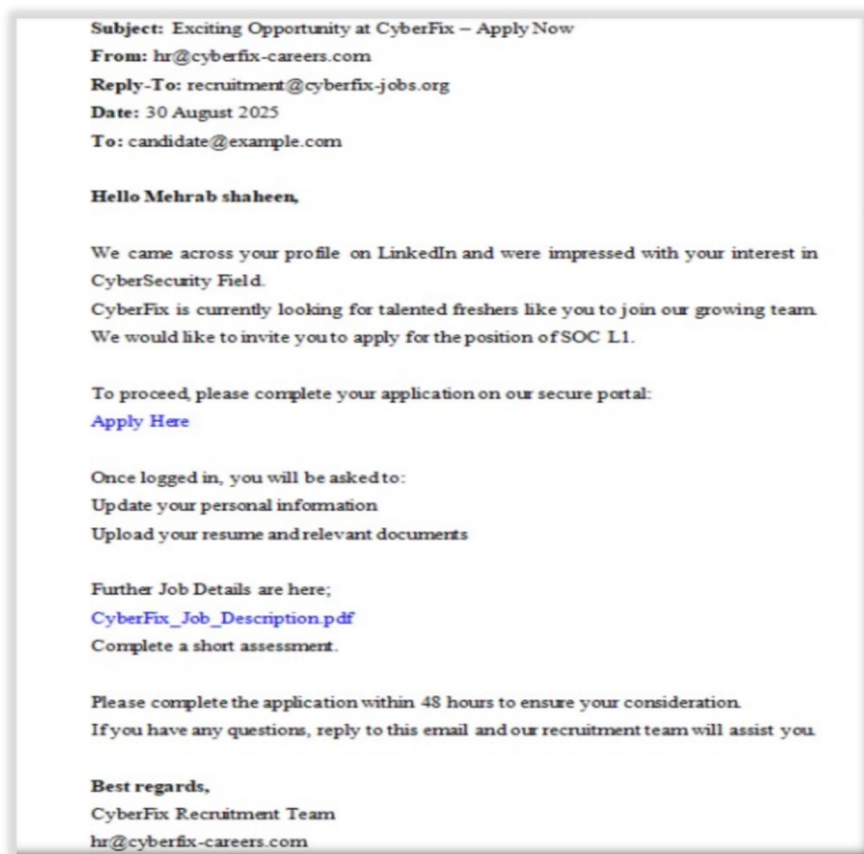


Exploring OpenCTI through a Phishing Campaign

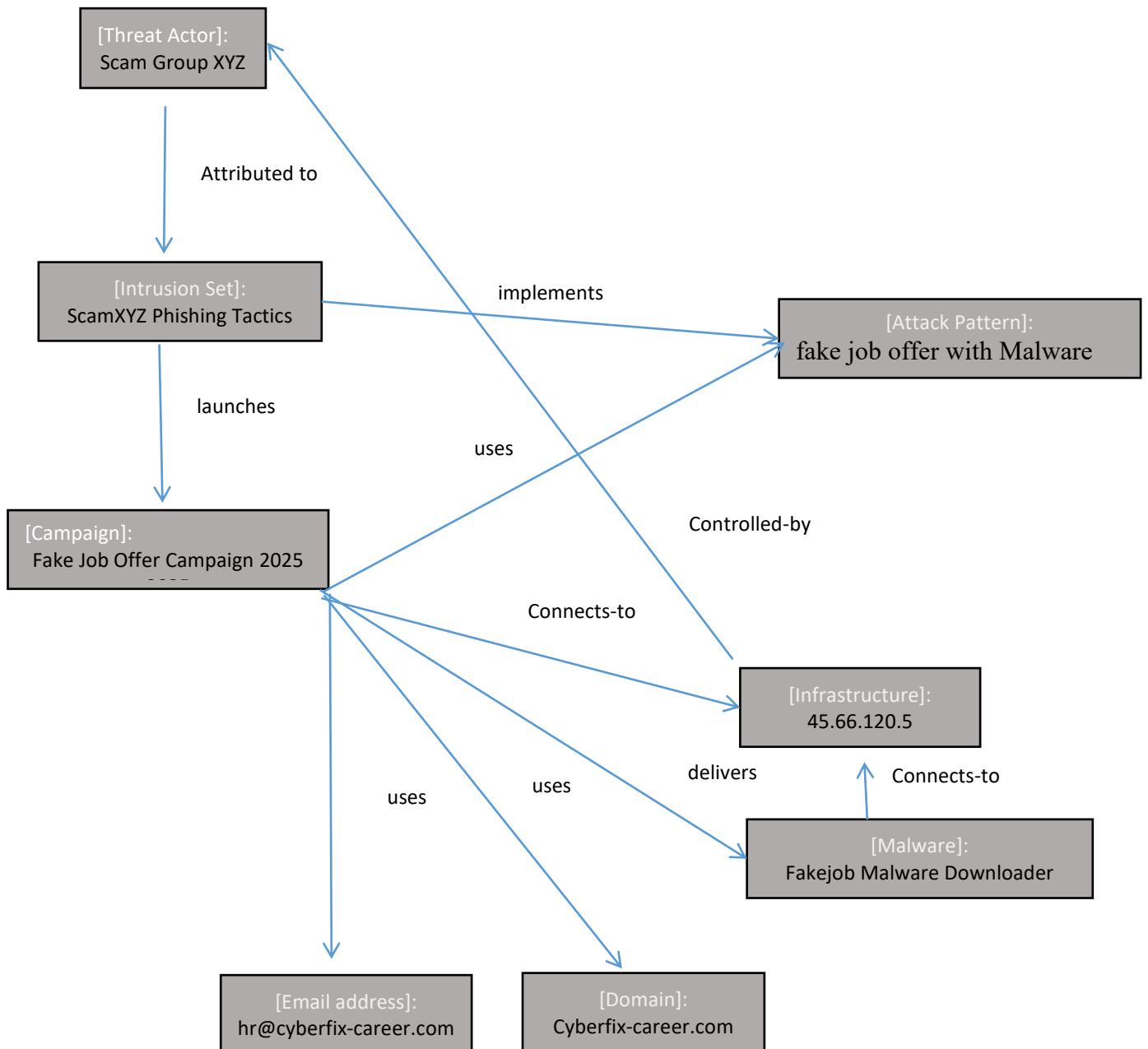
Executive Summary

A threat actor identified as “**Scam Group XYZ**” is conducting the “**Fake Job Offer Campaign 2025**” using the intrusion set “**ScamXYZ Phishing Tactics**”. The campaign impersonates the company **CyberFix**, sending fake job offer emails with a malicious PDF attachment. When opened, the payload executes and connects to a remote attacker-controlled IP, aiming to compromise victims’ devices and damage **CyberFix**’s reputation.

Phishing Email Targeting LinkedIn Users:



Overview of CTI for Phishing campaign scenerio:

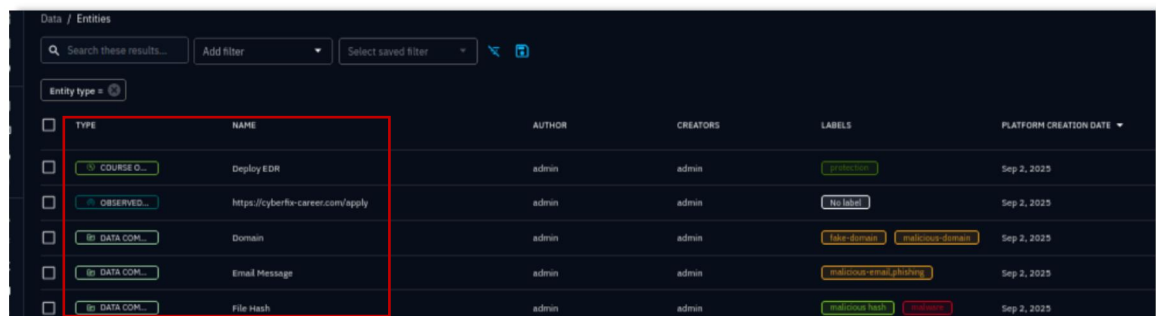


1. Entities

Objects in OpenCTI representing actors, campaigns, malware, observables, indicators, etc. It serves as structured representations of threats in the attack chain.

Examples:

- **Threat Actor:** ScamGroupXYZ
- **Campaign:** Fake Job Offer Campaign 2025
- **Intrusion Set:** ScamXYZ Phishing Tactics
- **Indicator:** Malicious email from hr@cyberfix-career.com



TYPE	NAME	AUTHOR	CREATORS	LABELS	PLATFORM CREATION DATE
COURSE OF ACTION	Deploy EDR	admin	admin	protection	Sep 2, 2025
OBSERVED	https://cyberfix-career.com/apply	admin	admin	No label	Sep 2, 2025
DATA.COM	Domain	admin	admin	fake-domain, malicious-domain	Sep 2, 2025
DATA.COM	Email Message	admin	admin	malicious-email-phishing	Sep 2, 2025
DATA.COM	File Hash	admin	admin	malicious-hash, malware	Sep 2, 2025

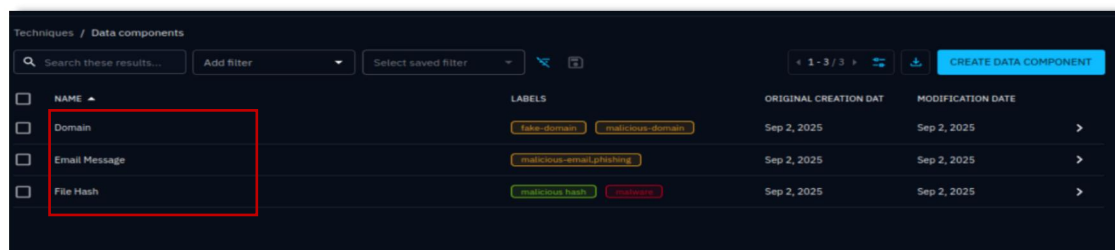
Entities represent each significant component of the attack, allowing analysts to map relationships and understand the overall threat landscape.

2. Data Components

Individual pieces of information that make up threat intelligence like emails, file hashes, IP addresses, URLs, malware samples. It represents raw or structured data collected during the investigation.

Examples:

- **Observable:** hr@cyberfix-career.com (email)
- **Artifact:** fakejob_offer.pdf



NAME	LABELS	ORIGINAL CREATION DATE	MODIFICATION DATE
Domain	fake-domain, malicious-domain	Sep 2, 2025	Sep 2, 2025
Email Message	malicious-email-phishing	Sep 2, 2025	Sep 2, 2025
File Hash	malicious-hash, malware	Sep 2, 2025	Sep 2, 2025

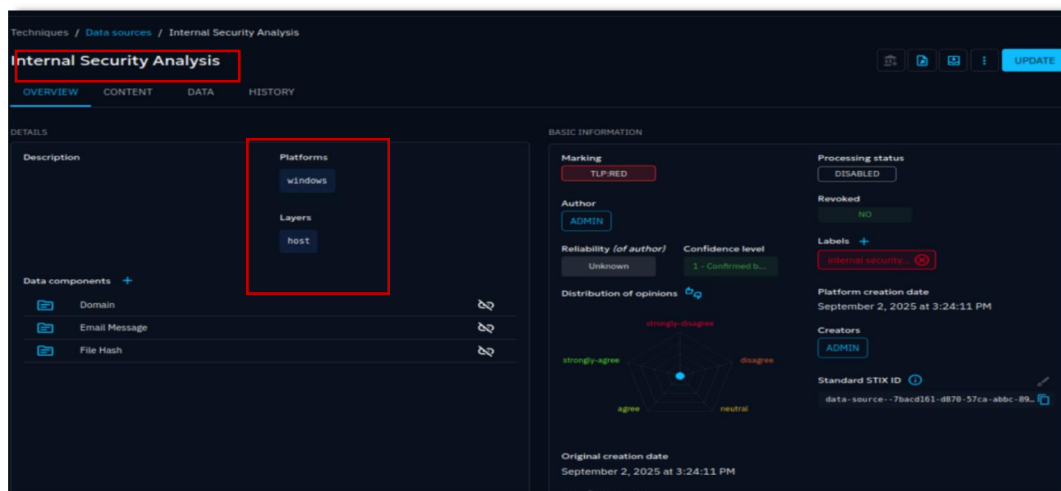
Data components are the building blocks of analysis, providing the foundational evidence from which indicators and attack patterns are derived.

3. Data Sources

It means from where the threat intelligence information comes. It establishes credibility and traceability of the collected data.

Examples:

- **Internal logs:** corporate email servers, SIEM, endpoint logs.
- **External feeds:** OSINT reports, phishing databases (PhishTank), threat intelligence platforms.
- **Social media:** LinkedIn monitoring for suspicious messages.



Data sources indicate the origin of each data component, enabling validation and correlation within the investigative workflow.

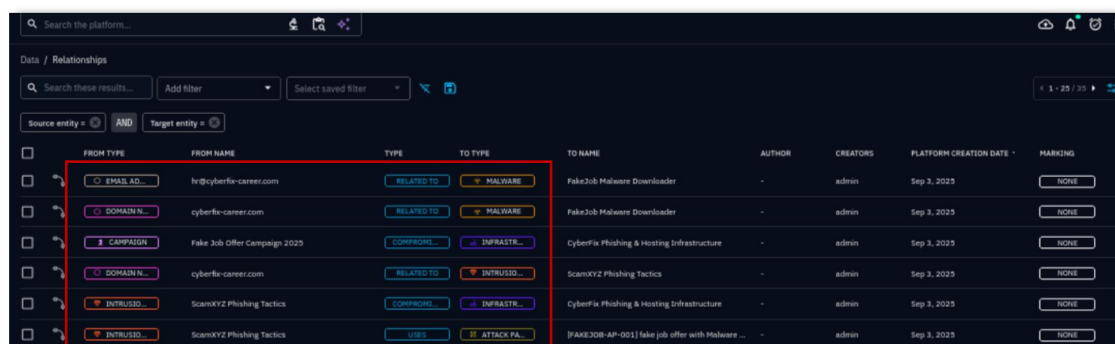
4. Relationships

Logical links connecting entities to illustrate the attack chain. It shows how different parts of the attack are related, supporting visualization and correlation.

Common Relationship Types:

based-on → Indicator derived from Observable

delivered-by → Artifact delivered by Campaign

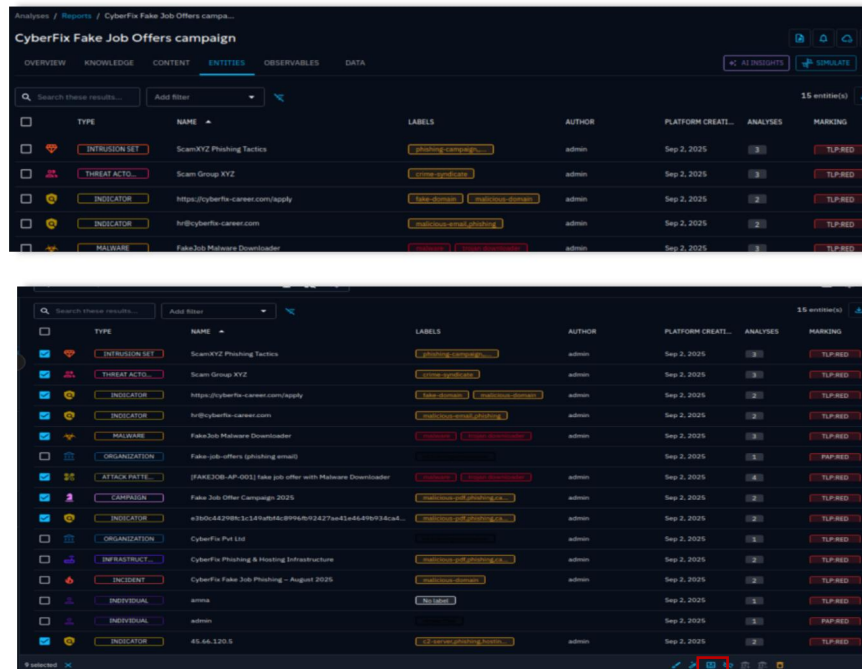


How to build and represent the relationships through graph in report:

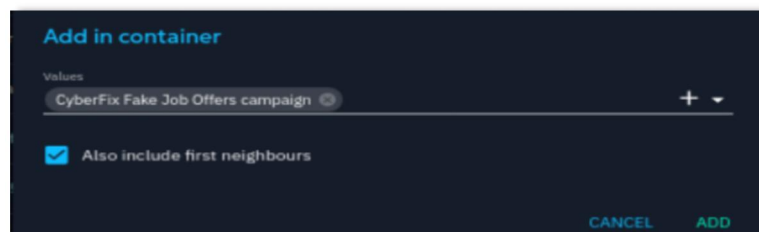
1. Create a new container like report.

Give it a title, description, and tags (TLP, confidence, etc.).

Go to > Entities



Select desired entities that are required to visualize CTI. Press Icon mentioned in image.



All selected entities are added in container “CyberFix Fake Offers campaign”.

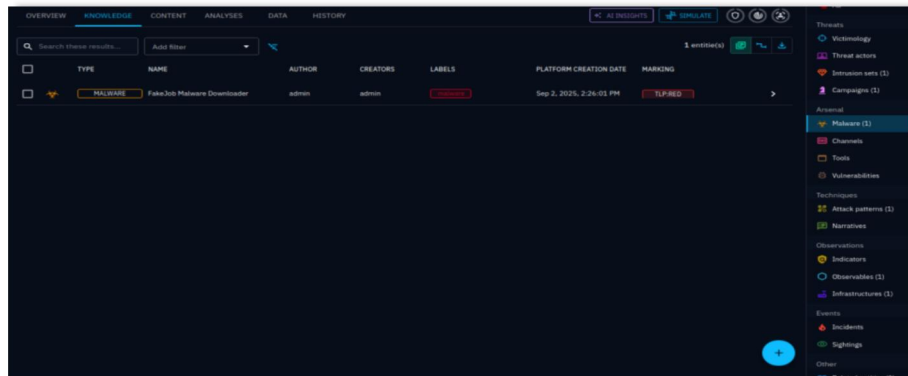
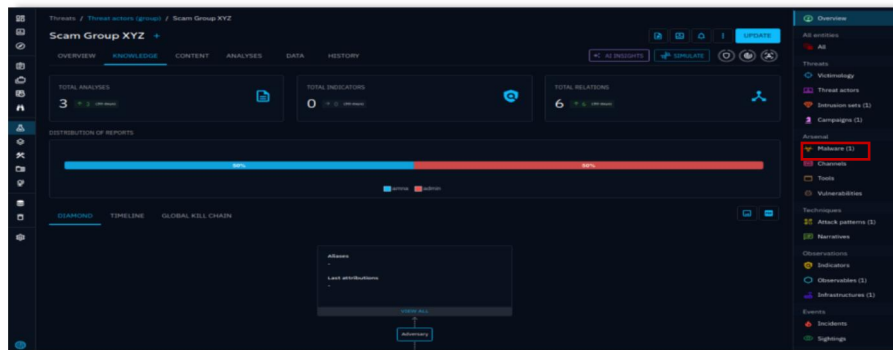
Note: Entities must be created before making relationships.

2. Build Relationships Between Entities

Go to the entity type like “Threat actor” for linking another entity type.

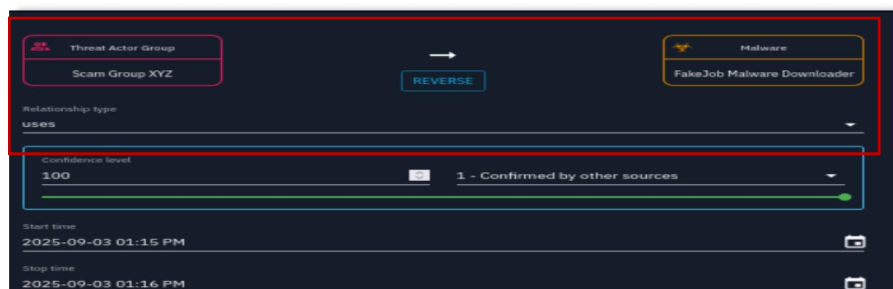
Go to > KNOWLEGDE option

Select any entity type that we want to link for relationship.



Press (+) icon.

Click continue.



Select required relationship type.

Note: Build relationships between entities and entities must be added into any container to represent a final graph for CTI.

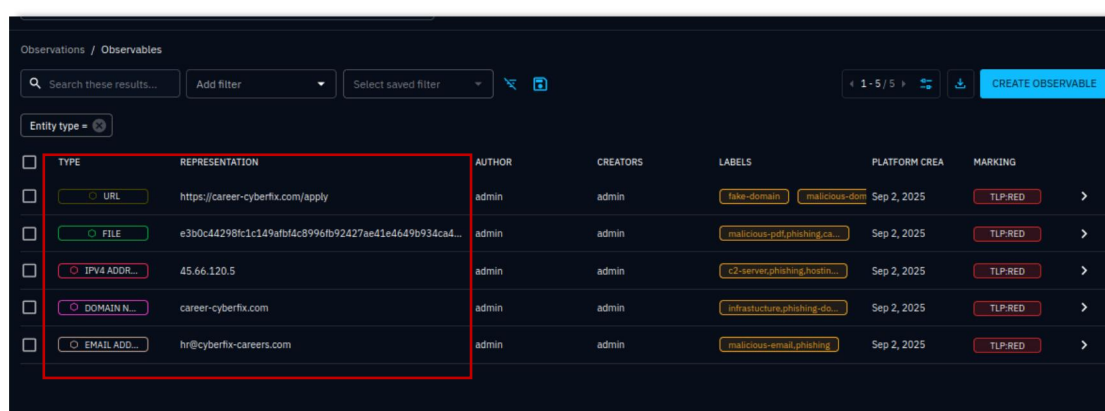
5. Observable Object

Purpose:

Observables are raw data points collected from internal or external sources, such as emails, domains, URLs, IPs, or file hashes, that indicate potentially malicious activity. They serve as the foundation of threat intelligence analysis, enabling the creation of indicators and linking to campaigns, malware, or threat actors.

Data source:

Observables are sourced from SIEM logs, EDR tools, firewalls, email servers, threat feeds, OSINT, and sandbox reports.



TYPE	REPRESENTATION	AUTHOR	CREATORS	LABELS	PLATFORM CREA	MARKING
URL	https://career-cyberfix.com/apply	admin	admin	fake-domain, malicious-dot	Sep 2, 2025	TLP:RED
FILE	e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca4...	admin	admin	malicious-pdf-phishing-ca...	Sep 2, 2025	TLP:RED
IPV4 ADDR...	45.66.120.5	admin	admin	c2-server-phishing-hostin...	Sep 2, 2025	TLP:RED
DOMAIN N...	career-cyberfix.com	admin	admin	infrastructure-phishing-do...	Sep 2, 2025	TLP:RED
EMAIL ADD...	hr@cyberfix-careers.com	admin	admin	malicious-email-phishing	Sep 2, 2025	TLP:RED

Scenario Context:

- Sender email address: **hr@cyberfix-career.com**
- Fake domain: **cyberfix-career.com**
- Hash of malicious PDF
- IP of fake domain.

These formed the initial evidence to analyze and track the threat.

Relationships:

Observables such as the phishing email, fake domain, and malicious PDF attachment are initially linked to indicators to represent actionable malicious activity.

- Observables → related-to → Indicators

Later, these indicators and observables will connect to artifacts, malware, and the campaign, forming the basis of the broader attack chain.

- Observable → related to → Artifact / Malware / Campaign

6. Indicator Object

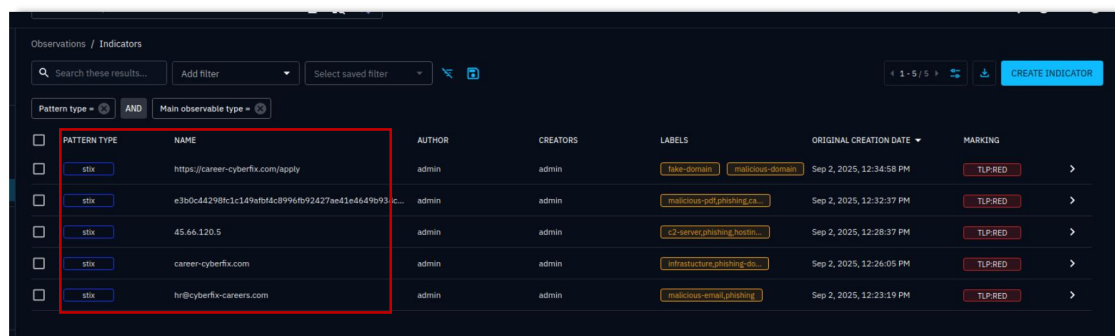
Purpose:

Indicators are actionable representations of suspicious or malicious activity derived from observables, such as emails, domains, file hashes, IPs, or URLs. They provide a way to detect, monitor, and respond to threats by enabling security tools like SIEM, IDS/IPS, and firewalls to generate alerts or block malicious behavior.

Data Source:

Indicators are created after analyzing observables to determine malicious intent and can be sourced from internal logs, threat intelligence feeds, OSINT, or sandbox analyses.

They also help correlate threats to campaigns, malware, and attack patterns, supporting effective incident response and proactive defense.



☐	PATTERN TYPE	NAME	AUTHOR	CREATORS	LABELS	ORIGINAL CREATION DATE	MARKING	
☐	stix	https://career-cyberfix.com/apply	admin	admin	fake-domain, malicious-domain	Sep 2, 2025, 12:34:58 PM	TLP:RED	>
☐	stix	e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b93c...	admin	admin	malicious-pdf-phishing-cs	Sep 2, 2025, 12:32:37 PM	TLP:RED	>
☐	stix	45.66.120.5	admin	admin	C2-server-phishing-hosts...	Sep 2, 2025, 12:28:37 PM	TLP:RED	>
☐	stix	career-cyberfix.com	admin	admin	infrastructure-phishing-ds	Sep 2, 2025, 12:26:05 PM	TLP:RED	>
☐	stix	hr@cyberfix-careers.com	admin	admin	malicious-email-phishing	Sep 2, 2025, 12:23:19 PM	TLP:RED	>

Scenario Context:

In the “Fake Job Offer Campaign 2025”, indicators created from the phishing email fake domain ,ip of fake domain, and malicious PDF to identify and track the threat.

Relationships:

Currently, indicators are derived from observables and **currently linked to them**.

- Indicators → derived-from → Observables

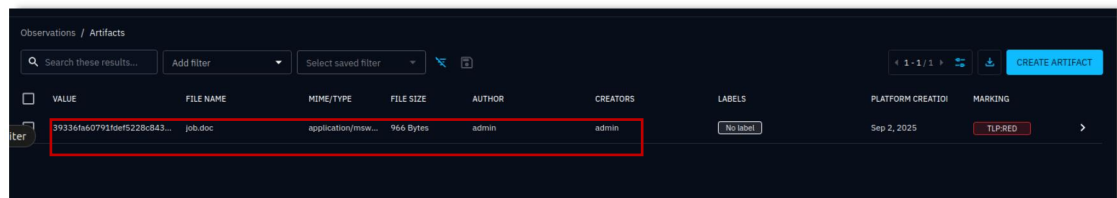
Later, they will be connect to artifacts, malware, and the campaign to provide context for detection and monitoring.

- Indicators → will later link → Artifact
- Indicators →will later link→ Malware

7. Artifact Object

Purpose:

Digital evidence like files, scripts, or malware samples used in attacks. It help to analyze the behavior, payload, and delivery method of threats.



VALUE	FILE NAME	MIME/TYPE	FILE SIZE	AUTHOR	CREATORS	LABELS	PLATFORM CREATION	MARKING
39336fa607910ef5228c843...	job.doc	application/msw...	966 Bytes	admin	admin	No label	Sep 2, 2025	TLP:RED

Scenario Context:

- In the “Fake Job Offer Campaign 2025”, the PDF attachment is considered an artifact because it contained the malicious payload used to compromise victims’ devices.
- The artifact is added to OpenCTI to correlate with indicators, malware, and the campaign, enabling monitoring and response actions.

Relationships:

Artifacts (malicious PDF) are delivered-by the campaign and linked to malware for payload execution.

- Artifact → delivered-by → Campaign
- Artifact → contains → Malware
- They will be connected to infrastructure components like attacker-controlled IPs.
- Artifact → uses → Infrastructure

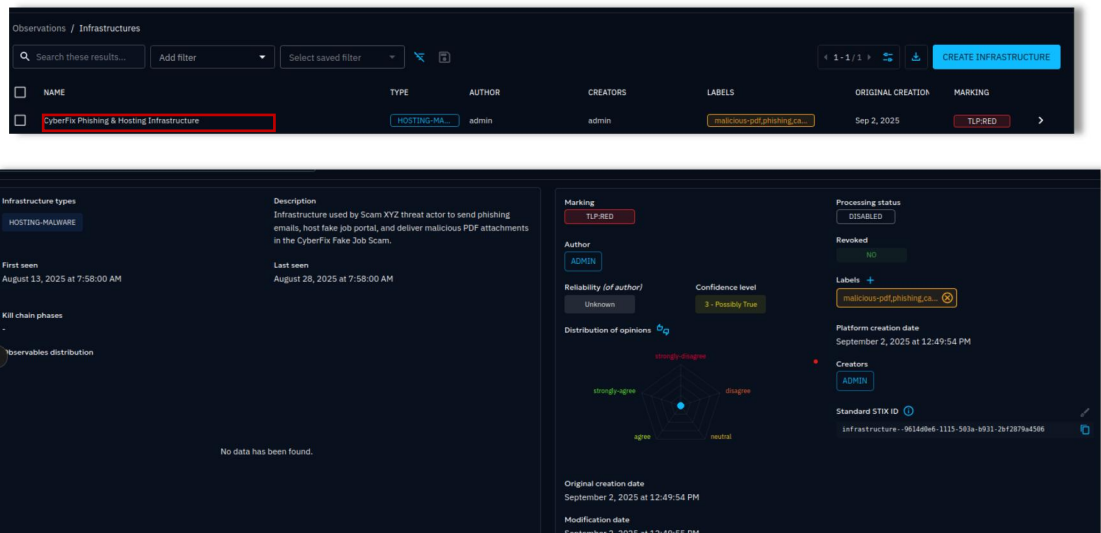
8. Infrastructure Object

Purpose:

Represent network components or resources used by attackers, such as C2 servers, malicious domains, or IP addresses. It help to map attacker activity and delivery infrastructure.

Usage:

Map attack infrastructure for tracking and mitigation. And link artifacts, malware, and campaigns to attacker resources.



Scenario Context:

- The attacker-controlled IP that the PDF payload connected to is considered infrastructure, as it enabled remote access and control of compromised devices.
- In OpenCTI, the infrastructure entity is linked to the artifact and malware, showing the connection between the PDF and the attacker-controlled system.

Relationships:

Infrastructure (attacker IP) is currently linked to malware and artifacts to show operational resources.

- Infrastructure → supports → Artifact / Malware

Later, it provides context for the threat actor's activities.

- Infrastructure → later context → Threat Actor

9. Malware Object

Purpose:

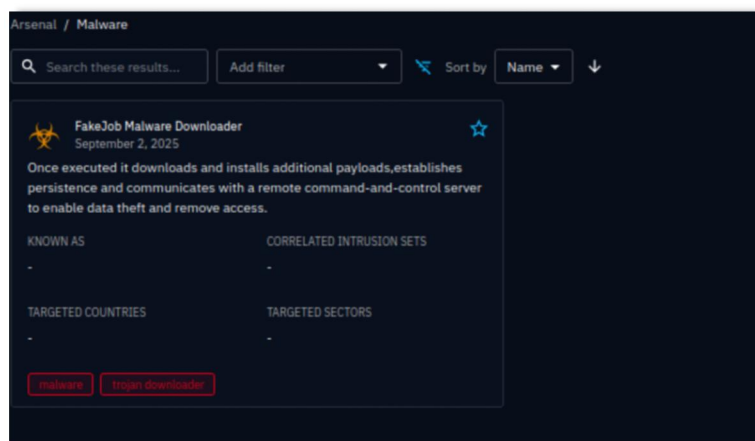
Represents the malicious payload or software used in attacks. It helps to analyze attack behavior, techniques, and potential impact.

Data Source:

Sandbox analysis, endpoint telemetry, malware repositories, threat intelligence feeds.

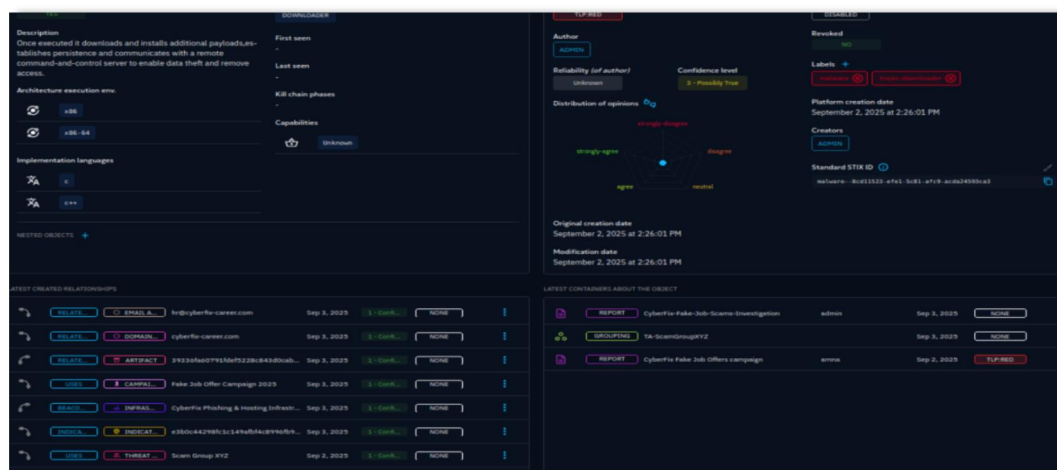
Usage:

Link to indicators, artifacts, campaigns, and attack patterns. And develop detection rules and mitigation strategies.



Scenario Context:

- The FakeJob Malware Downloader executed from the PDF attachment to compromise victims' devices in the "Fake Job Offer Campaign 2025."
- Malware is identified through sandbox execution of the PDF and correlated with threat feeds.



- The malware entity in OpenCTI is linked to the artifact (PDF), infrastructure (attacker IP), and campaign.



Relationships:

Malware is linked to artifacts and implements attack patterns. It is also connected to the campaign and later to the intrusion set and threat actor.

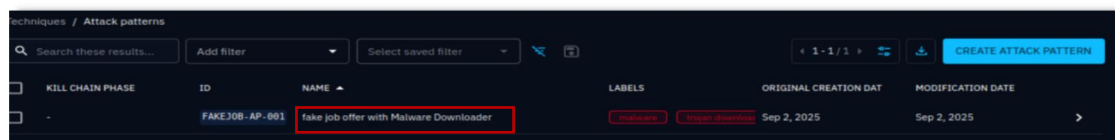
- Malware → contained-in → Artifact
- Malware → implements → Attack Pattern
- Malware → part-of → Campaign
- Malware → later link → Intrusion Set / Threat Actor

10. Attack Pattern Object

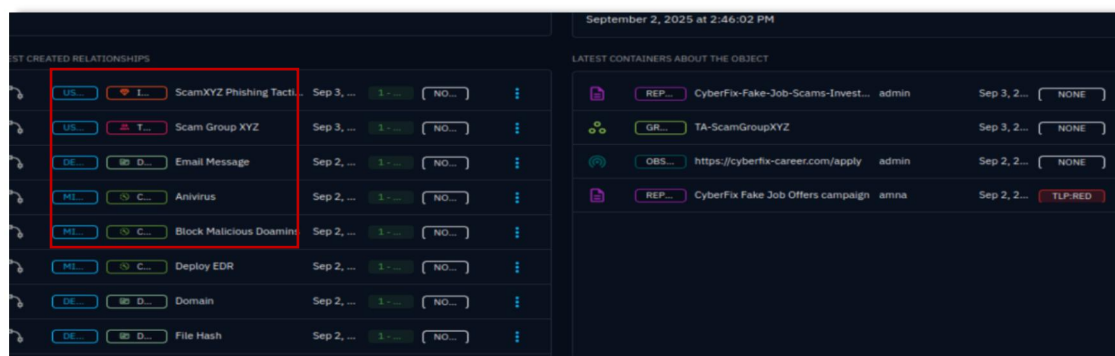
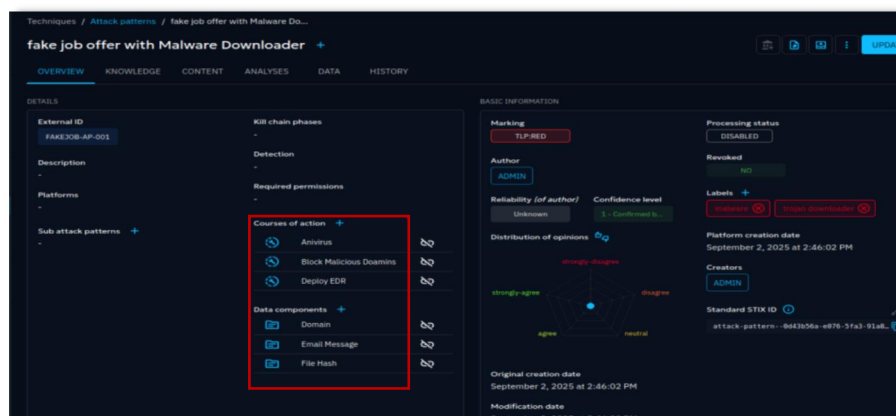
Purpose:

Represent the techniques or tactics used by attackers (e.g., MITRE ATT&CK).

Provide insight into “how” the attack is/was executed.



Scenario Context: “fake job offer with Malware Downloader” is identified as the attack pattern for the campaign.





Relationships:

Attack patterns represent techniques used by malware or campaigns. They are linked to malware and indicators.

- Attack Pattern → linked-to → Malware / Indicator

Later, it will be linked to campaigns, intrusion sets, and threat actors.

- Attack Pattern → part-of → Campaign → Intrusion Set → Threat Actor

11. Campaign Object

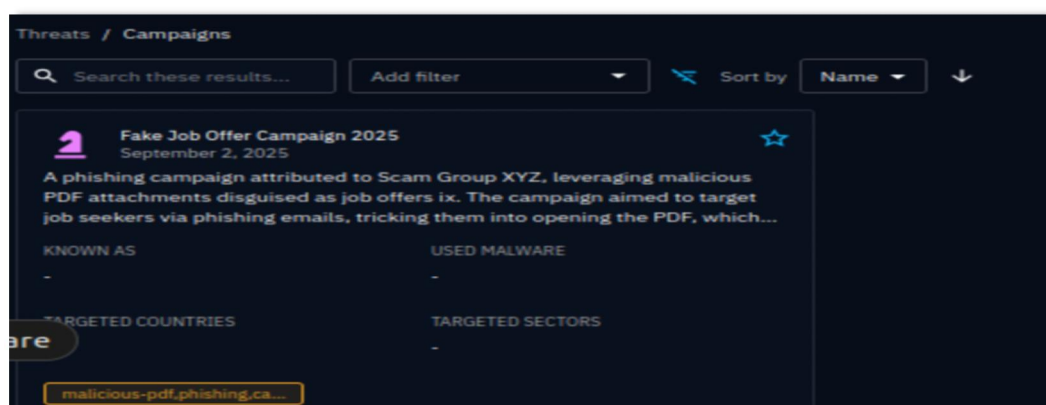
Purpose:

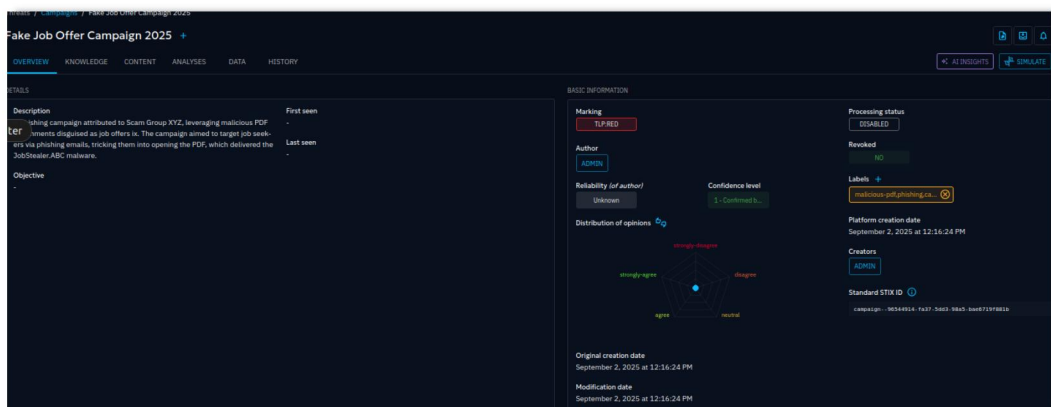
Represents a coordinated set of malicious activities with a specific objective.

Provides context for linking indicators, artifacts, malware, and attack patterns

Data Source:

Observables, indicators, malware analysis, threat intelligence feeds, user reports.





Scenario Context:

- “Fake Job Offer Campaign 2025” is launched to impersonate CyberFix and deliver malicious PDFs to victims.
- Email logs, threat feeds, and sandbox results helped identify and define the campaign.
- Campaign entity in OpenCTI linked all related observables, indicators, artifacts, and malware.

Relationships:

Campaign aggregates related activity, linking observables, indicators, artifacts, and malware.

- Campaign → uses → Indicators / Malware / Artifacts

It is later part of the intrusion set and associated with the threat actor.

- Campaign → part-of → Intrusion Set
- Campaign → attributed-to → Threat Actor

12. Intrusion Set Object

Purpose:

Group campaigns that share similar tactics, techniques, and procedures (TTPs).

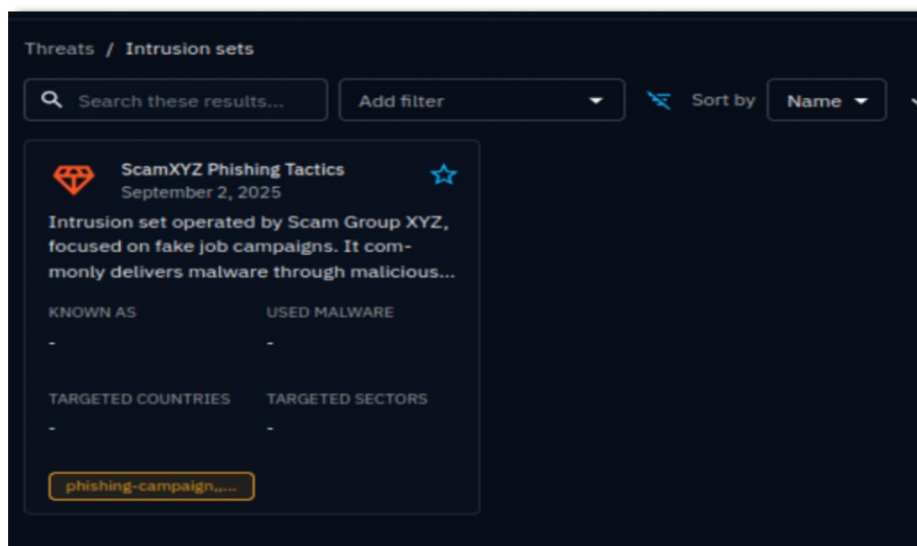
Identify broader patterns of threat actor behavior.

Data Source:

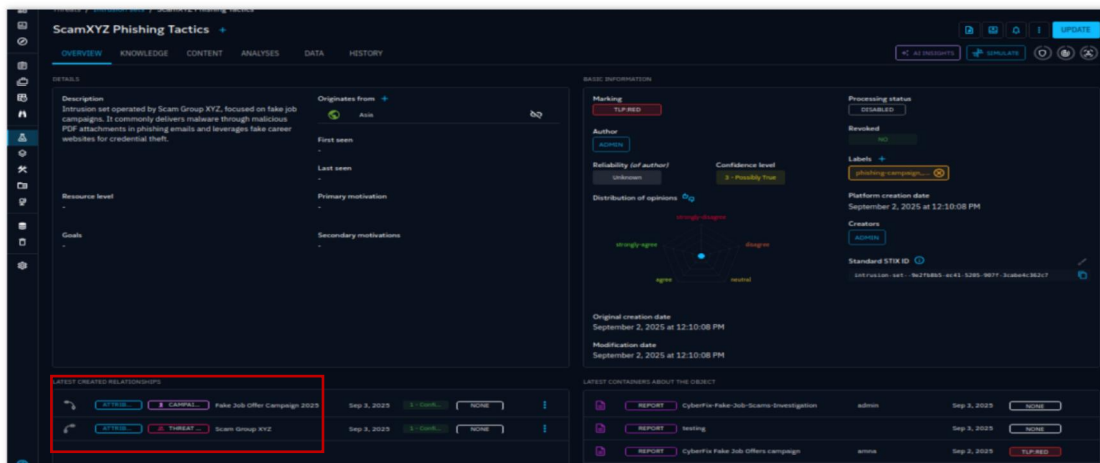
Historical campaigns, threat intelligence feeds, OSINT, internal security reports.

Usage:

Understand threat actor methods and track long-term activity. It links campaigns to specific threat actors for attribution.



“ScamXYZ Phishing Tactics” grouped multiple campaigns based on similar phishing methods.



Campaign similarities and TTP analysis are used to define the intrusion set.



Scenario Context:

- Intrusion set linked campaigns and helped associate the activity to “Scam Group XYZ”.

Relationships:

Intrusion set groups multiple campaigns with similar TTPs and is linked to campaigns. It is ultimately attributed to the threat actor.”

- Intrusion Set → groups → Campaign
- Intrusion Set → attributed-to → Threat Actor

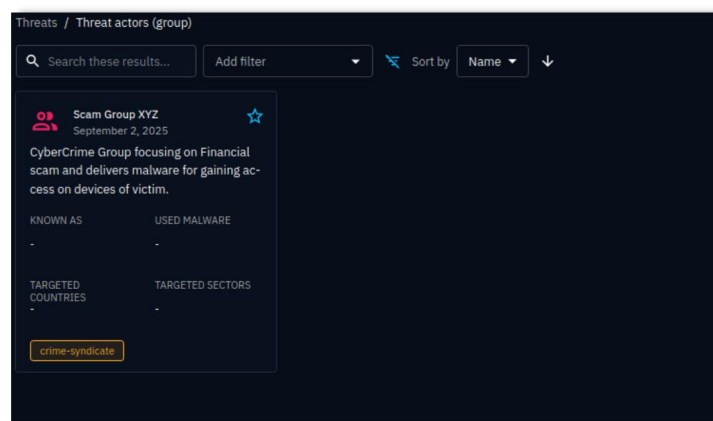
13. Threat Actor Object

Purpose:

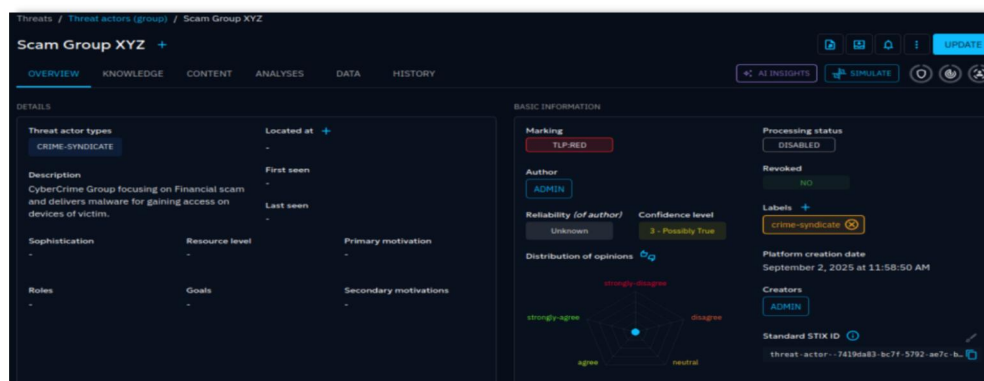
Represents the individual, group, or organization behind malicious activity.

Usage:

Connect campaigns, intrusion sets, and attack patterns to an actor. It supports mitigation, threat hunting, and reporting.



“ScamGroupXYZ” is identified as the threat actor responsible for the “Fake Job Offer Campaign 2025”.



TEST CREATED RELATIONSHIPS				Modification date September 2, 2025 at 11:58:50 AM	
US	1	CyberFix Phishing & Ho...	Sep 3, ...	3	NO...
RE	D	cyberfix-career.com	Sep 3, ...	3	NO...
AT	1	ScamXYZ Phishing Tacti...	Sep 3, ...	3	NO...
PA	C	Fake Job Offer Campaig...	Sep 3, ...	3	NO...
US	A	[FAKEJOB-AP-001] fak...	Sep 3, ...	3	NO...
US	M	FakeJob Malware Down...	Sep 2, ...	3	NO...

LATEST CONTAINERS ABOUT THE OBJECT					
REP	CyberFix-Fake-Job-Scams-Invest...	admin	Sep 3, 2...	NONE	
GR	TA-ScamGroupXYZ		Sep 3, 2...	NONE	
REP	CyberFix Fake Job Offers campaign	amina	Sep 2, 2...	TLP:RED	



Scenario Context:

- In OpenCTI, the threat actor entity is linked to the intrusion set, campaign, malware, and attack patterns to complete the intelligence picture.

Relationships:

Threat actor is responsible for the observed attack. It is linked to the intrusion set and indirectly to all campaigns, malware, artifacts, indicators, and observables.”

- Threat Actor → responsible-for → Intrusion Set → Campaign → Malware / Artifact / Indicator → Observable
- Threat Actor → controls → Infrastructure

14. Courses Of Action

These are actions or mitigations taken to respond to or prevent a threat. It shows how security teams can contain, mitigate, or remediate malicious activity.

Examples:

- Blocking a malicious URL or IP.
- Quarantining malicious files or attachments.
- Updating firewall or endpoint detection rules.

Techniques / Courses of action

Search these results... Add filter Select saved filter 1 - 3 / 3 CREATE COURSE OF ACTION

NAME	LABELS	ORIGINAL CREATION DAT	MODIFICATION DATE
Anivirus	anivirus	Sep 2, 2025	Sep 2, 2025
Block Malicious Doamins	Block-malicious-domains	Sep 2, 2025	Sep 2, 2025
Deploy EDR	edp-protectos	Sep 2, 2025	Sep 2, 2025

“Courses of Action” outline the specific steps taken or recommended to prevent further compromise, based on analysis of indicators, artifacts, and threat actor behavior.

15. Report Object

A **“Report”** in OpenCTI is a container that groups related threat intelligence entities, observables, and relationships into one documented case. It acts like a threat intelligence report that analysts or organizations create to describe a campaign, incident, or threat activity.

CyberFix Fake Job Offers campaign

OVERVIEW KNOWLEDGE CONTENT ENTITIES OBSERVABLES DATA

Description

Write Preview

Content

Paragraph

This report documents a malicious campaign where threat actors distributed fake job offers via email to lure potential victims. The phishing emails contained malicious attachments and links that, when executed, installed a Trojan malware on the victim's system. The malware was designed to establish persistence, exfiltrate sensitive information, and potentially allow remote access by the attacker.

The campaign demonstrates the combination of social engineering tactics and malware delivery, targeting individuals searching for employment opportunities. This activity highlights an ongoing trend of using job-related lures to achieve initial access in targeted environments.

Findings / Key Observations:

Threat actors used phishing emails with the subject line "New Career Opportunity" to trick victims.

Malicious attachments contained a Trojan (Remote Access Trojan family).

ENTITY DETAILS

Description

Report types

THREAT-REPORT

Publication date

September 2, 2025 at 1:09:33 PM

Correlated containers

REPORT CyberFix-Fake... admin Sep 3, 2025 NONE

Entities distribution

Indicator

Individual

Organization

Artifact

Attack Pattern

Campaign

Domain name

Email address

Incident

Infrastructure

0 1 5

BASIC INFORMATION

Marking

TLP:RED

Author

AMNA

Reliability (of author)

Unknown

Confidence level

3 - Possibly True

Distribution of opinions

strongly-disagree

strongly-agree

agree

neutral

disagree

Processing status

NEW

Assignees

ADMIN

Participants

ADMIN

Revoked

NO

Labels

report

Platform creation date

September 2, 2025 at 1:15:25

Creators

ADMIN

Original creation date

September 2, 2025 at 1:09:33 PM

Modification date

September 3, 2025 at 10:30:29 AM

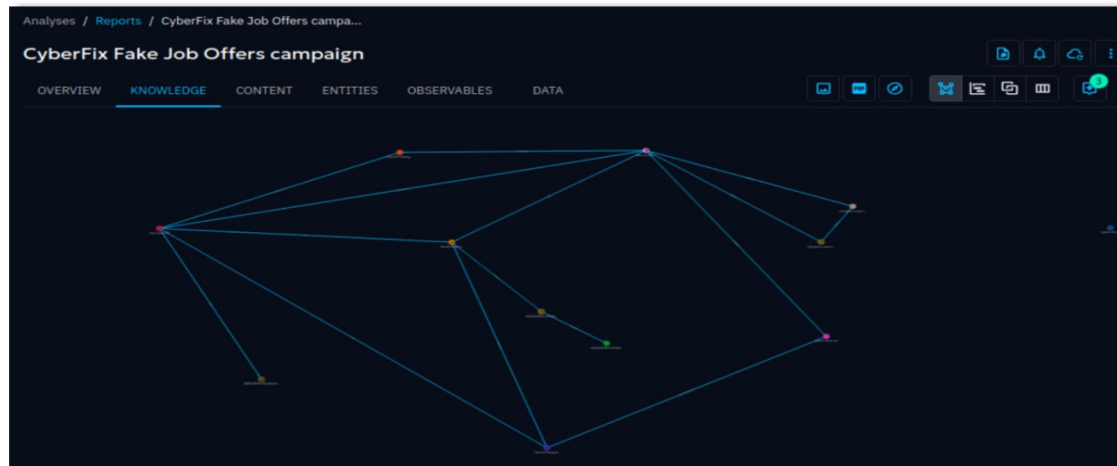
Standard STIX ID

report--c0a2db8c5-fe7d-58f4-e

I created report to analyze and track the evidences of Fake Phishing campaign.

Report Graph

Visual summary of the report showing entities and their relationships. It allows readers to see the attack chain at a glance.



The report graph visually represents the attack chain for the “Fake Job Offer Campaign 2025”.

Nodes: Represent entities (Observable, Indicator, Artifact, Malware, Campaign, Intrusion Set, Threat Actor, Infrastructure).

Edges: Represent relationships such as **Campaign** → **attributed-to** → **Threat Actor**.

Links show how indicators, malware, and campaigns ultimately relate to the threat actor. This graph helps analysts quickly identify key points for mitigation.

16. Home Tab

High-level overview of all incidents and campaigns, showing how multiple campaigns, intrusion sets, threat actors, and infrastructure are related for monitoring and trend analysis.

